Name: ____________________________

Date: ______________________________

Mr. Rawson • Y9 MYP Design

3a.2 – How the Game Lab Grid Works

Unit 3a: Interactive Animations & Games

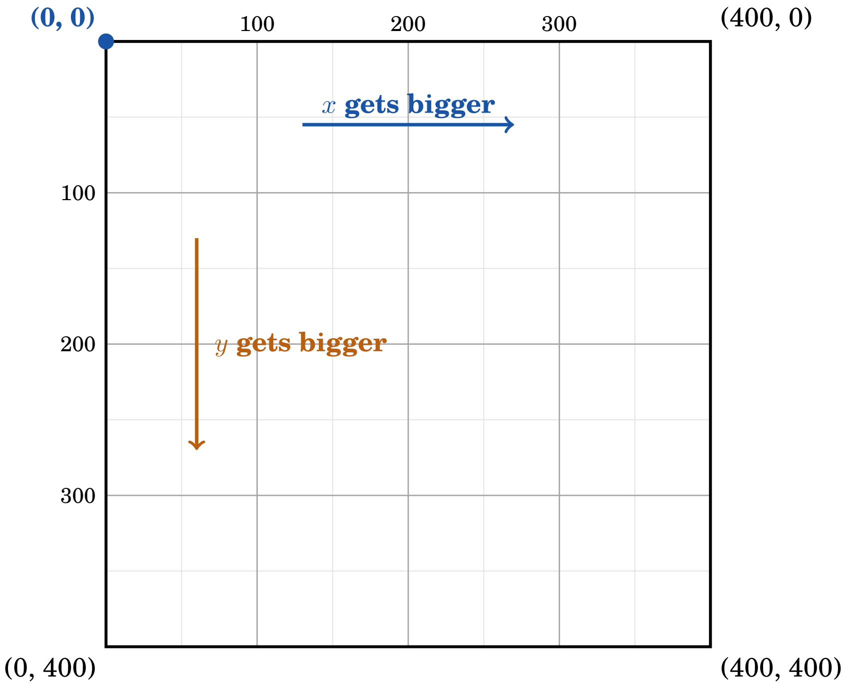
Learning intentions

By the end of this activity you can:

- say where the origin is on the Game Lab grid, and which way y increases;
- give a partner instructions precise enough to reproduce a drawing you cannot see;
- explain why the *order* of your instructions changes the picture.

1 The canvas is 400 by 400

Everything you draw in Game Lab goes inside a fixed square window, **400 pixels wide and 400 pixels tall**. It does not scroll and it does not grow. If you send a shape to 500, it is off the edge and you will not see it.



The three rules of the grid

1. The window is **400 across and 400 down**.
2. **$(0, 0)$ is the TOP-LEFT corner** — not the middle, not the bottom.
3. x counts **right**. y counts **DOWN**.

Why is it upside down?

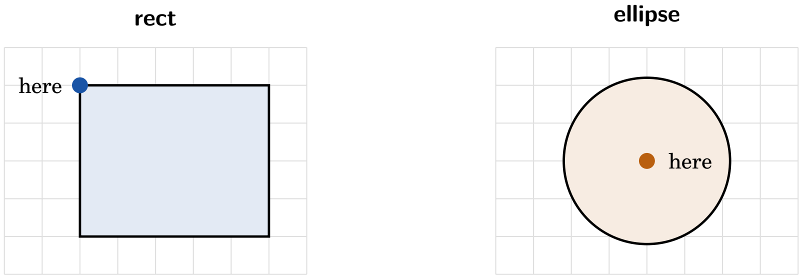
In maths, y goes *up* and the origin sits in the middle, so half your coordinates are negative. Screens do it differently for two good reasons:

- We **read** from the top left, and a page grows *downwards* as you add to it. So a screen starts counting where the reading starts.
- It means **every coordinate is a positive number**. There are no negatives to worry about anywhere on the canvas.

This is the one that catches people. A bigger y means *further down the screen*, not further up. If your shape appears at the wrong end of the canvas, this is almost always why.

2 Where exactly does the shape sit?

Every shape has one **black dot** — the point that lands on the coordinate you give. **It is not in the same place on every shape**, and this is the second thing that goes wrong.



a **rectangle** is placed by its **top-left corner**

a **circle** is placed by its **centre**

So "put a circle at 200, 200" and "put a square at 200, 200" do *not* line up the same way. Say which shape you mean, and remember where its dot lives.

3 Colour and order

- **Colour is set before the shape is drawn**, and it stays until you change it. If you forget to change it, the next shape comes out in the last colour you used.
- **Later shapes cover earlier ones.** The order you give your instructions in *is* the order they stack. Background first, details last.

4 Check yourself before you start

1. Which corner of the canvas is $(0, 0)$? ____________
2. A shape moves from $y = 100$ to $y = 300$. Has it moved up or down? ____________
3. Give the coordinates of the **bottom-right** corner of the canvas. ____________
4. You draw a red square, then a blue circle on top of it. Which instruction did you give *first*? ____________
5. Your partner says "circle at 200, 200". Where on the canvas is the *centre* of that circle? ____________
6. Why does a computer screen put $(0, 0)$ at the top left instead of the middle?

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Activity

Now do the activity. You will get a **Version A** or **Version B** sheet — *do not show it to your partner*. Take turns: one of you describes the drawing, the other uses the Game Lab tool to build it without looking at the sheet. Keep this page beside you.